

Communicating Meaning and Team Role Through Gesturing Telepresence Robots

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Abstract—We ran two studies to explore how gesturing telepresence robots can support better communication in distributed teams. Crowdsourced participants provided their impressions of: (1) the meaning of individual gestures, and (2) the social roles of team partners. Collaborators were better understood when their robotic intermediaries portrayed gestures in concert with their facial expressions. Also, teammates on both sides of the interaction were perceived as more involved in the conversation, more composed in demeanor, and more equal in stature.

Keywords—telepresence; robotics; embodiment; collaboration.

I. GESTURE AND ROBOTIC AVATARS

Most of us live in a world of words. Yet we communicate with each other in ways that draw upon a cultural language of behavior that frequently takes place outside of our conscious awareness [1]. This language includes behaviors such as gesture and movement, posture, physical proximity and body orientation, eye gaze, facial expressions and even non-verbal vocalizations. We draw upon these actions every day, without even thinking about it, to ease our abilities to think improvisationally and to speak expressively.

Until very recently, video has been the only way to convey such non-verbal information between each other at a distance. But video chat does not provide the full breadth of rich, gestural content that we exchange in our collocated interactions, because the images typically only display one's head and shoulders. And while conferences capture a greater range of behaviors, distant collaborators become literally and figuratively two-dimensional. That is, they cannot move around the physical space where their partners are located, so their presence may readily be forgotten. They cannot reach through their screens to touch people or objects within that space, so their ability to collaborate is compromised.

We have been developing physically embodied, remotely actuated, robotic avatars for use during distributed collaboration. These hybrids draw upon the advantages of both live video, which communicates facial expressions and vocal characteristics, and robotics, which manifest an embodied, physical presence. The devices allow team members who cannot be physically present to extend themselves into the group's workspace. Remote participants control robotic components that represent their heads, necks and arms—using motion-sensitive interfaces—so that they can look around, gesture expressively and point at shared artifacts. They can



Figure 1. As the remote collaborator leans forward to look closer, his robot avatar gestures in concert. Study participants reported that complementary onscreen and physical actions were more understandable and had greater impact on them than the facial expression or robotic motion alone.

therefore be considered *physical avatars*, personifications that represent particular individuals in another place.

II. STUDIES IN EXPRESSION AND ROLE

We are employing these avatars in two related studies that explore the role of avatar-enabled non-verbal communication in distributed design activities. The first study focuses on the basics of how people interpret actions presented by a remote participant who is communicating by robotic avatar during a conversation. The second study shifts focus to how people infer the personalities and roles of remote and local participants whom they observe interacting in a team setting.

Study 1: Interpreting Avatar Action

We created video prototypes combining onscreen and robotic actions that we wanted to explore, and crowdsourced study participants to interpret them using Amazon's Mechanical Turk. A video prototype is a brief movie clip that demonstrates how an interactive technology would perform [3], and provides many benefits over live, in-person trials. It permits us: to tune the robot's motions to be as subtle or obvious, coarse or refined as we intend; to reproduce precise timings between the remote collaborator's onscreen gestures and the robot's motions; to repeat the exact same scenario(s) to expose to our study participants, for as many times as the study requires (which was more than a hundred in our case, and the setup and breakdown of a lab study can be laborious and fraught with intermittent breakdowns); or to gain access to an audience that is more diverse in background—including age, culture, experience with conferencing technologies—than would be possible by recruiting only local participants.

We showed 164 participants nine gestures in three different forms: facial expressions alone, robotic motion alone, and combined expression and robotic motion (shown in Fig. 1). For the latter condition, we moved the avatar in concert with the onscreen action. We found that participants were more correct in their interpretations of the expression-plus-motion condition than they were for either of the other two conditions ($\chi^2(8, N=164) = 395.724, p < .001$) (see Fig. 2 for details). Participants were also more confident in their interpretations, and felt that the intended message had a greater impact upon them (shown below in Table 1).

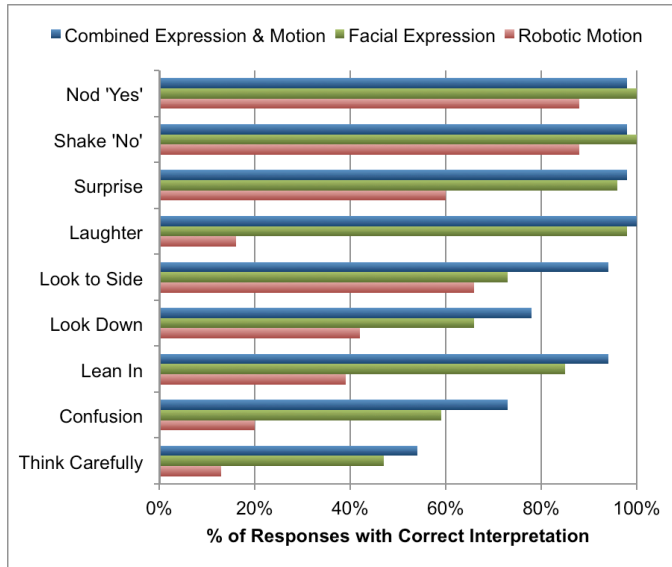


Figure 2. The percentage of responses that agreed with the intended message for participants located in the United States. *Laughter* was well recognized from viewing facial expressions alone; in contrast, *nod 'yes'* was well recognized from viewing the robotic motion alone.

	Confidence			Strength of Response		
	Comb	Fac Exp	Rob Mot	Comb	Fac Exp	Rob Mot
<i>M</i>	5.74	5.55	4.40	5.38	5.25	4.29
<i>SE</i>	0.51	0.57	0.46	0.45	0.49	0.29

Table 1. Participants' subjective ratings for confidence in their interpretations of the gestures shown, and their strength of response to those gestures, for Study 1.

However, their abilities to interpret meanings varied not only from one condition to another, but also from one gesture to another ($\chi^2(8, N=164) = 345.94, p < .001$). That is, certain forms of body language were more readily understandable than others (see Fig. 2 again). For example, nodding one's head is an outwardly communicative action, and participants recognized it readily, even when shown robotic motion alone. Laughter is more of an unexpected emotional response, and participants relied to a much greater degree on facial cues (we provided no audio) to recognize it. Motions that indicate one's focus of attention, such as looking over to one side were less well recognized overall, but combining facial expressions and robotic motion improved the situation significantly. Gazing



Figure 3. A remote collaborator interacts with his colleagues through a robot avatar. Using a motion-sensing interface, he can point at features of a prototype (top), humorously compare hand size (bottom), wave, move objects and even tap on the table to get someone's attention.

downward while confused or immersed in thought is a less intentional action; it reflects a more thoughtful, internal state, and participants typically had difficulty recognizing it.

Study 2: Role in a Team Setting

For our follow-up study, we also employed video prototypes. We created an interactive design session scenario between three colleagues, one of whom participated remotely through an avatar, and showed it to 103 study participants. We adapted the Relational Messaging Scale [3] as a measure of the social relations between the colleagues, and asked participants about their perceptions of the interaction. The scale describes the balance between two main indicators, dominance and affiliation, through a number of narrower concepts that are more readily observable.

We recorded four different versions of the design session, depicting the same scene, dialogue, participants and overall sequence of action (shown above in Fig. 3). The storyline had the remote collaborator asking his local teammates for help redesigning a remote control that they had recently designed together. We changed only two characteristics of the participants between versions, which we chose because of their relevance to the relational scale. The first was whether the remote collaborator's physical avatar moved along with his actions, or remained idle for the duration of the session. The second was the degree to which either the remote collaborator,

or the local teammate, took a dominant role in leading the session.

The addition of robotic motion over the idle condition led study participants to rate the remote collaborator as more friendly and involved than in the case where his avatar could not move (see Table 2). This makes sense, as the ability to enact body language influences one's own sense of involvement in a conversation. Participants also rated the local teammate as being more equal to her partner when he could perform such gestures.

Variable		M	SE	Sig	Eta ²
Remote's Friendliness	Idle avatar	4.53	1.03	.02 F(1,3)=5.71	.05
	Robot motion	5.05	1.24		
Remote's Dominance	Idle avatar	3.97	1.14	.055 F(1,3)=3.78	.04
	Robot motion	3.47	1.41		
Remote's Involvement	Idle avatar	4.98	1.28	.03 F(1,3)=5.00	.05
	Robot motion	5.50	1.36		

On-site's Equality	Idle avatar	4.50	1.24	.04 F(1,3)=4.20	.04
	Robot motion	4.95	1.50		

Table 2. Significant differences between whether the remote team member's avatar remained idle or displayed robotic motion during the design session.

Participants rated the remote collaborator as more composed and involved when he took a leadership role in the discussion, compared to when he took a follower role (see Table 3). But they also rated the local teammate as more composed and equal for the same scenario: when the remote collaborator led the discussion. These responses, along with the preceding finding of greater teammate equality during avatar motion, suggest that the addition of physical gestures serves as a moderating influence on the perception of *both* participants. One interpretation of this finding is that participating in a meeting through an avatar that does not move—which can be considered similar to current forms of video chat—introduces a sense of inequality between the partners; and this inequality can be ameliorated by re-introducing the non-verbal cues that embodied physical motion enables.

Variable		M	SE	Sig	Eta ²
Remote's Composure	Remote led	5.31	1.33	.02 F(1,3)=5.77	.05
	On-site led	4.72	1.33		
Remote's Involvement	Remote led	5.49	1.33	.04 F(1,3)=4.47	.04
	On-site led	5.00	1.32		

On-site's Composure	Remote led	5.29	1.30	.02 F(1,3)=5.34	.05
	On-site led	4.71	1.25		
On-site's Dominance	Remote led	3.12	1.32	.02 F(1,3)=6.31	.06
	On-site led	3.78	1.34		
On-site's Equality	Remote led	4.98	1.46	.03 F(1,3)=4.78	.05
	On-site led	4.47	1.27		

Table 3. Significant differences between whether the remote team member took a leader versus a follower role during the team interaction.

III. CONCLUSION

Taken together, the two studies suggest that gesture and motion in embodied telepresence setups favorably influence our ability to understand the meaning, as well as perceive the social role, of distant collaborators, and to feel confident in those perceptions. We are currently developing a more robust, field-ready version of the robotic avatar, both to run these studies with in-person participants and to explore how longer-term interactions may influence perceptions.

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REFERENCES

- [1] Hall, E. (1959) *The silent language*. Garden City, NY: Doubleday.
- [2] Ju, W. and Takayama, L. (2009) Approachability: How people interpret automatic door movement as gesture. *International Journal of Design* 3(2), 77-86.
- [3] Dillard, J., Solomon, D. and Palmer, M. Structuring the concept of relational communication. *Communication Monographs* 66, 49-65.